

SMX Taggant-Selection System – Final HLD

Resource Inventory

Purpose: Buildable Azure resource list for the AI-powered marker (taggant) selection system. Rationalized from the two source HLDs into a single YAGNI-compliant plan.

Build order: Dev spoke first; Prod spoke added at go-live. Hub built once, shared.

Primary design driver: *Correctness of the deliverable* (a wrong marker recommendation causes real-world harm) – this elevates retrieval grounding and observability to first-class concerns, not just ops.

Design principles (locked)

- Private-by-default** – all PaaS data/AI services reached over private endpoints; public egress is the single anonymizing Search Proxy only.
- Correctness over cleverness** – agents answer from *retrieved sources* (AI Search) and *deterministic lookups* (the element-limits workbook), never from model memory; every regulatory claim carries a citation.
- Fewer reasoning hops** – specialize agents by *judgment*, expose capabilities (search, list, retrieve) as *tools*. More agents = more hallucination surface.
- YAGNI** – no resource without a confirmed need; cost-heavy items deferred or flagged, not pre-provisioned.
- Dev lighter, Prod full** – same resource *types* across environments; dev runs lower tiers / relaxed posture, prod runs the hardened set. Upgrade paths are in-place where possible.

1. Hub VNet – Shared Services (built once)

Resource	Dev	Prod	Notes
Hub VNet	✓	✓	Shared-services hub
Application Gateway	Standard_v2 (no WAF, or WAF_v2 Detection)	WAF_v2, Prevention	Front door per HLD; dev skips the WAF premium
Public IPs	1 (AppGw frontend)	2 (per HLD)	Static

Resource	Dev	Prod	Notes
Private DNS Zones (<code>privatelink.*</code>)	✓	✓	Centralized resolution for all PEs
VNet Peerings (hub ↔ spoke)	✓	✓	
VPN / ExpressRoute Gateway	⚠ flag	⚠ flag	PROBABLY REDUNDANT — no on-prem connectivity identified. Kept on diagram; provision only if a hybrid need appears.

2. Spoke networking (per environment)

Resource	Dev	Prod	Notes
Spoke VNet	✓	at go-live	
Agent subnet (delegated <code>Microsoft.App/environments</code>)	✓	✓	Hosts the Container Apps environment
Private Endpoints subnet	✓	✓	Data & AI layer PEs
Azure Functions subnet	✓	✓	Isolates the public-egress proxy
NSGs (per subnet)	✓	✓	Standard hygiene

3. Compute layer

Resource	Dev	Prod	Notes
Azure Container Apps environment (workload profiles)	Consumption profile	Dedicated (D4) — <i>size to real load</i>	Hosts all containers below
→ Frontend (UI) container	✓	✓	Same ACA env, behind App Gateway
→ Backend API container	✓	✓	Orchestration entry point
→ Orchestrator + Agent containers (Intake & Discovery, Chemistry, Compliance)	✓	✓	Self-hosted ; specialized capabilities are <i>tools</i> , not extra agents

Resource	Dev	Prod	Notes
Microsoft Foundry account + project	✓	✓	Model inference + platform tools via Responses API only
→ Model deployment (GPT-4o-class)	✓	✓	Reasoning model for agents
→ Embedding model deployment (e.g. text-embedding-3-large)	✓	✓	For push-based index ingestion
Azure Functions app (isolated)	✓	✓	Search Proxy (HTTP, anonymized public egress – justified by IP/NDA posture) + Monthly Regulatory Sync (timer trigger)
Container Registry	Standard (public + firewall)	Premium (private endpoint)	Premium required for PE; dev runs lighter

Cut from compute: Foundry **Capability Host / Agent Service runtime** – redundant once orchestration is self-hosted; we manage conversation state in our own Cosmos. Also removes the agent-subnet Service Association Link (the main private-networking teardown headache).

4. Data layer (Medallion)

Resource	Dev	Prod	Notes
ADLS Gen2 – Bronze	✓	✓	Raw XRF spectra, MSDS/regulatory PDFs. Hot tier; lifecycle → Cool as cost lever. Est. few hundred MB → ~1-2 GB
Cosmos DB (NoSQL) – Silver/Gold	Serverless	Serverless (→ autoscale only if sustained load)	Refined rules, marker library, project records, chat threads, recommendation audit records (rationale + citations as first-class docs) . Data is small; throughput is the only cost driver. No 3,000 RU/s floor (Capability Host dropped)
Azure AI Search	Basic (~\$75/mo)	S1 (~\$250/mo)	Push-based ingestion (sync function chunks + embeds + pushes). Corpus ~150-300 MB indexed – tiny. Basic→S1 is an in-place upgrade. S1's value is concurrency + private integrated-vectorization, which only prod needs

AI Search tier is driven by networking + ingestion model, not data size. Push ingestion keeps Basic viable under a private endpoint (no outbound private links needed).

5. AI services

Resource	Dev	Prod	Notes
Microsoft Foundry	✓	✓	(see Compute)
Azure AI Speech	✓	✓	Voice in/out for the consultative, revision-heavy human-in-the-loop UX
Azure AI Content Safety	✗	✗	CUT – "dangerous" here = <i>incorrect taggant</i> , not toxic content. Nothing in the source files requires moderation

6. Security & Ops

Resource	Dev	Prod	Notes
Log Analytics Workspace	✓ (central, in hub)	✓ (same workspace)	Both spokes feed one workspace. Retention = cost lever
Application Insights (workspace-based)	✓	✓	Distributed trace across orchestrator → agents → tools = the "why" behind each recommendation. Doubles as Foundry's required App Insights connection
Azure Monitor (alerts/dashboards)	✓	✓	
Key Vault	Standard, platform-managed keys	Standard, platform-managed keys	Holds residual secrets (proxy API key, webhook secrets). Encryption-at-rest already on by default
Customer-Managed Keys (CMK)	⚠ flag	⚠ flag	DROPPED unless mandate confirmed (see open items). Would also bump KV → Premium
Managed Identities (SMI + UMI)	✓	✓	Auth backbone – secret-less; keeps KV minimal
Defender for Cloud – free CSPM	✓	✓	Free secure-score everywhere
Defender paid plans (Containers, Storage, KV)	✗	✓ select	Per-resource cost; prod-hardening flip
Microsoft Sentinel (SIEM)	⚠ flag	⚠ flag	PROVISION ONLY IF a SOC or compliance mandate is confirmed. A SIEM nobody triages is pure cost. Kept on diagram, marked

7. Private Endpoints & DNS (per environment)

Private endpoint + matching `privatelink` DNS zone for each:

Key Vault · Microsoft Foundry · Azure AI Search · Cosmos DB · ADLS Gen2 (blob + dfs) · Azure AI Speech · Container Registry (*Prod*)

Cut list (removed, with reason)

Item	Why cut
Foundry Capability Host / Agent Service runtime	Redundant with self-hosted orchestration; removes SAL complexity
Azure AI Content Safety	Wrong threat model – correctness ≠ content moderation
CMK on Key Vault	No mandate confirmed; platform-managed encryption is free and on by default
WAF in Dev	Prod-hardening step, not a dev need
2nd Public IP in Dev	Single front-end IP sufficient for dev
Cosmos provisioned 3,000 RU/s floor	Was a Capability Host requirement – gone with it

Flagged (kept visible, provision on trigger)

Item	Trigger to provision
VPN / ExpressRoute Gateway	A real hybrid / on-prem connectivity need appears
Microsoft Sentinel	SMX confirms an active SOC or a compliance/contractual SIEM requirement
CMK (+ Premium KV)	A customer-managed-key mandate (ISO 27001 / SOC 2 / client contract) is confirmed
Prod ACA Dedicated D4 sizing	Real prod load measured

Open items to confirm (owner: whoever holds SMX compliance/ops)

1. **SOC / compliance status** – ISO 27001, SOC 2, data residency, or client CMK clauses? → settles Sentinel + CMK.
2. **Hybrid connectivity** – any future on-prem / instrument link? → settles the Gateway.
3. **Prod concurrency** – expected concurrent users / throughput? → settles Cosmos serverless-vs-provisioned and ACA prod sizing.